Hardware Dimensions

ESP32-S3 voice assistant prototype | 27 October 2026 | All sizes in millimetres (mm)

How these were measured: XIAO pin positions come from Seeed's official KiCad footprint files (exact). Module sizes come from manufacturer listings. Values marked **(photo)** were estimated from a photo using the breadboard as a ruler and are only accurate to about +/-15%. Clone modules vary by seller: confirm with vernier calipers before designing a PCB or enclosure.

1. Overall module sizes

Part	Size (L x W x H)	Notes
XIAO ESP32-S3	21.0 x 17.8 mm	Height not published; about 4-5 mm with USB-C (measure)
INMP441 mic module	about 14 mm round	Sellers list 14-15 mm
MAX98357A amp module	19.4 x 17.8 x 3.0 mm	Adafruit original; purple clones similar (measure)
Speaker (box type)	about 36 x 17 mm body, 49 mm with ears (photo)	Looks like a common 2040 type (40 x 20 mm). Depth about 6-8 mm
Breadboard (400 holes)	82.5 x 55 x 8.5 mm	Used as the measuring scale

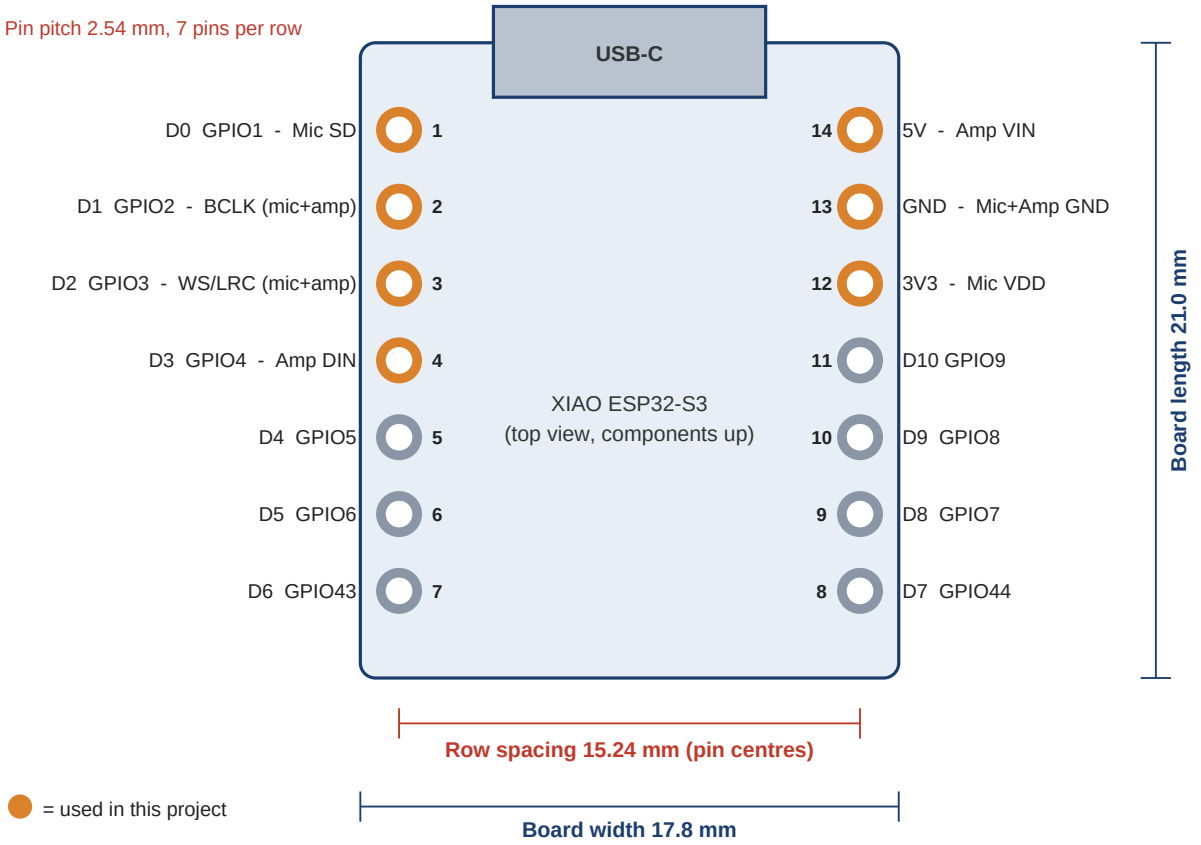
2. Pin and row dimensions

Part	Pins	Pitch	Row spacing	Other
XIAO ESP32-S3	14 (2 rows x 7)	2.54 mm	15.24 mm	Drill 0.889 mm, pad 1.524 mm; row span 15.24 mm
INMP441 module	6 (2 cols x 3)	2.54 mm	7.62 mm	Pin order varies by seller
MAX98357A module	7 (1 row)	2.54 mm	single row	Row length 15.24 mm; speaker terminal usually 3.5 mm pitch
Breadboard	-	2.54 mm	7.62 mm centre gap	30 columns x 10 rows + 4 power rails
Speaker	2 wires	-	-	Measure mounting-hole spacing

3. XIAO ESP32-S3 footprint details

Feature	Size	Position / spacing
Through-hole pins (DIP)	Pad 1.524 mm, drill 0.889 mm	Rows 15.24 mm apart, 2.54 mm pitch
Castellated edge pads	2.432 x 1.524 mm	Pad centres 16.91 mm apart across the board
Surface-mount (SMD) pads	2.75 x 2.0 mm	Rows 16.165 mm apart (pad centres)
Back pads (8 extra signals)	1.7 mm round	2 cols x 4 rows on a 2.54 mm grid
Back pads (likely battery + / -)	2.5 x 1.1 mm	1.905 mm apart
Back pad (likely ground / thermal)	2.75 x 2.5 mm	Under the ESP32-S3 chip

4. XIAO pin layout and project connections



Drawn to scale (4x), viewed from the component side with USB-C at the top. Pins 1-7 (D0-D6) run down the left side from the USB end; pins 8-14 run back up the right side, ending at 5V next to the USB port.

5. Wiring used in this project

Signal	XIAO pin	INMP441 mic	MAX98357A amp
Bit clock	D1 (GPIO2)	SCK	BCLK
Word select	D2 (GPIO3)	WS	LRC
Mic data	D0 (GPIO1)	SD	-
Speaker data	D3 (GPIO4)	-	DIN
Power	3V3 / 5V	VDD to 3V3	VIN to 5V
Ground	GND	GND, and L/R to GND	GND
Not connected	-	-	GAIN, SD

6. Chip sizes (for a future custom PCB)

A custom board needs only the chips, not the modules. These are the parts inside (or replacing) the modules above.

Function	Chip	Package size
Processor + Wi-Fi + Bluetooth	ESP32-S3R8 (8 MB PSRAM inside)	7 x 7 mm (QFN56)
Flash memory (8 MB)	GigaDevice GD25Q64	about 3 x 4 mm (USON) to 5 x 6 mm (SOP8)
Crystal	40 MHz	2.0 x 1.6 mm

Function	Chip	Package size
Antenna	Chip antenna	about 3.2 x 1.6 mm + copper-free area
Microphone (current)	INMP441 (not for new designs)	4.72 x 3.76 x 1.0 mm
Microphone (replacement)	ICS-43434 (same I2S signals)	3.50 x 2.65 x 0.98 mm
Amplifier	MAX98357A	3 x 3 mm (TQFN) or 1.35 x 1.44 mm (WLP)
Ready-made module option	ESP32-S3-WROOM-1 N8R8	18 x 25.5 x 3.1 mm

7. Official files and sources

- Seeed XIAO KiCad footprints: files.seeedstudio.com/wiki/XIAO-KiCad-Library/New_XIAO_Series_Footprints.zip
- XIAO ESP32-S3 dimension drawing (DXF): files.seeedstudio.com/wiki/SeeedStudio-XIAO-ESP32S3/res/XIAO_ESP32S3_v1.1_Dimensioning.dxf
- XIAO ESP32-S3 wiki: wiki.seeedstudio.com/xiao_esp32s3_getting_started
- Adafruit MAX98357A breakout: adafruit.com/product/3006
- TDK INMP441: invensense.tdk.com/digital/inmp441